HEALTH-BAND Neuro — Production Datasheet

EoS Health · Model EOS-HBN-001 · Revision 1.0

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1. Product Overview

HEALTH-BAND Neuro is a wristband combining 8-channel surface electromyography (sEMG), electrodermal activity (EDA/GSR), 3-lead ECG, PPG, and therapeutic TENS neurostimulation in a single device. It is the only wearable that simultaneously monitors muscle activation patterns and delivers targeted pain relief and recovery stimulation.

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2. Electrical Specifications

Parameter	Min	Typ	Max	Unit
Battery voltage (Li-Po)	3.0	3.7	4.2	V
Battery capacity	—	300	—	mAh
Active current (sEMG + ECG + BLE)	—	38.5	48	mA
TENS stimulation current (max)	—	—	20	mA
TENS pulse charge (max)	—	—	3.0	µC
Deep sleep current	—	8.1	12	µA
Wireless charging (Qi)	—	5W	—	—
Charging time (0→100%)	—	50	70	min

3. Sensor Suite

Sensor	IC	Measurement	Accuracy	Sampling Rate
sEMG (8-channel)	ADS1299	Muscle activation, fatigue index	SNR ≥72 dB, noise <0.5 µV_rms	1000 Hz
ECG (3-lead)	ADS1292R	Lead I/II/III, HR, HRV, AFib	SNR ≥63 dB	500 Hz
EDA/GSR	AFE4490	Skin conductance, stress index	0.01–100 µS	64 Hz
PPG	MAX30102	SpO_2, HR, HRV	ARMS ≤0.44%	100 Hz
Accelerometer/Gyro	LSM6DSO	6-DOF motion, gesture	±0.1 g	104 Hz
Temperature	MCP9808	Skin temperature	±0.25°C	1 Hz
TENS output	MAX14521E	Biphasic stimulation, 4 channels	±20 mA, 1–150 Hz	Configurable

4. Wireless Specifications

Parameter	Value
BLE version	5.2
PHY	1M / 2M
Typical range	100 m
Charging	Qi wireless (5W), WPC 1.3 compliant
Antenna	PCB trace antenna (integrated in flex PCB strap)

5. Mechanical Specifications

Parameter	Value
Core module dimensions	45 × 35 × 11 mm
Strap type	Flexible PCB silicone strap (8 sEMG electrodes embedded)
Strap sizes	S (140–170 mm), M (170–200 mm), L (200–230 mm)
Weight	42 g (medium strap)
Housing material	Polycarbonate + TPU overmold
IP rating	IP68 (1 m, 30 min)
Operating temperature	−10°C to +50°C
Flex strap fatigue life	≥500,000 flex cycles (IEC 60068-2-21)

6. Processor and Memory

Component	Part	Specification
MCU	Nordic nRF52840	Cortex-M4F, 64 MHz, 1 MB Flash, 256 KB RAM
AI co-processor	MAX32666	Dual Cortex-M4F, TFLite Micro, 1 MB Flash
NVM	W25Q64JV	8 MB SPI Flash
Secure element	ATECC608B	Key storage, OTA verification

7. Power Budget

Mode	Current	Duration	Energy
Active (sEMG + ECG + BLE)	38.5 mA	5 min/hr	3.208 mAh/hr
TENS therapy	52.0 mA	20 min/hr	17.333 mAh/hr
Idle (BLE + IMU)	5.1 mA	30 min/hr	2.550 mAh/hr
Deep sleep	8.1 µA	5 min/hr	0.001 mAh/hr
Weighted average	**~56.5 mA**	—	—
Battery life	—	**~5.3 days**	—

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8. TENS Safety Specifications

HEALTH-BAND Neuro TENS output complies with IEC 60601-2-10 (nerve and muscle stimulators):

Parameter	Specification
Waveform	Symmetric biphasic rectangular
Pulse width	50–400 µs (user configurable)
Frequency	1–150 Hz (user configurable)
Peak current	0–20 mA (user configurable)
Max charge per pulse	3.0 µC (IEC 60601-2-10 limit: 50 µC)
Optical isolation	Yes (ISO7241C between MCU and output stage)
Overcurrent protection	Hardware comparator, <1 µs response
Contraindications	Pacemaker, pregnancy, epilepsy, open wounds

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9. BOM Summary (Production, 1k units)

Category	Key Components	Unit Cost
MCU + AI	nRF52840, MAX32666, ATECC608B	$9.40
Sensors	ADS1299, ADS1292R, AFE4490, MAX30102, LSM6DSO	$14.20
TENS	MAX14521E, optical isolators, output capacitors	$5.80
Power	BQ25185 PMIC, Qi receiver, 300 mAh Li-Po	$6.20
Flex PCB strap	8-electrode FPCB, silicone overmold	$11.50
Housing + assembly	PC/TPU housing, packaging	$9.80
Total BOM		**$56.90**
Target retail		**$349**